

Mar 18, 2026

CUDA:OpenCL translation - Transcript

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Samyak Jhaveri: Um there are five kinds of a transforms and total 240 results uh from the augmentation. Now the augmentation is being done by Errol. He's really helping me out really fast with uh augmentations named L excuse me L0 to L4 and they are essentially increasingly um aggressive augmentations and increasing aggressive changes to the code itself so that to obiscate they're translating LLM from knowing what or from guessing from memory what that kernel might be. So essentially we are uh asking it to not rely on its memory but rely on

Gal Oren: H.

Samyak Jhaveri: its reasoning skills and that's essentially what we want to look in research in LLM's uh for parallel code generation moving forward.

Gal Oren: Mhm.

Samyak Jhaveri: Um I have been designing the LLM evaluation um

Gal Oren: Excuse

Samyak Jhaveri: harness as well uh since the last couple days. We have a lots of issues with the augmentation code because they were very focused and driven

Gal Oren: me.

Samyak Jhaveri: for OpenMP and they were making a lot of errors for CUDA and for other APIs. So we had to spend a lot of time especially errors spend a lot of time working on that and fixing that.

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Samyak Jhaveri: Um there's a distribution of the kinds of kernels we have from physics to machine

Gal Oren: Yeah.

Samyak Jhaveri: learning all the way to image graph crypto and sorting.

Gal Oren: Mhm.

Samyak Jhaveri: It is a 4070 with 12 GB of VRAM. So it's good enough to have a baseline but not the best for ideal experiments. Um the third is augmentation itself which is Tom to Tom Tom and Errol's

Gal Oren: Mhm.

Samyak Jhaveri: code. And then finally we are evaluating LLMs on their ability to translate from one API to the other. And as of now, just for testing purposes, we have done it for 240 CUDA to open MP and Open CL pairs. Uh it's not 240 each, it's 240 total. And right now, uh this is the breakdown of where it skipped, where it failed and so on, um across it,

Gal Oren: mostly green.

Samyak Jhaveri: uh for Hecbench.

Gal Oren: Mhm.

Samyak Jhaveri: So it passes for CUDA. it does not pass for OM and 60 kernels are tested so far. This is for the baseline uh evaluation.

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Gal Oren: Mhm.

Samyak Jhaveri: Then there's cross cross API pilot.

Gal Oren: Mhm.

Samyak Jhaveri: We did it for 20 Rodinia kernels and um a very dismal pass rate. U it all failed because I don't have AMD GPU um

Gal Oren: Mhm.

Samyak Jhaveri: it managed to do so.

Gal Oren: Mhm.

Samyak Jhaveri: the individual build times for all the kernels. This is exceptionally long. I'm sure it's like a huge GPU compute uh pro thing.

Gal Oren: Mhm. Mhm.

Samyak Jhaveri: Um he Yeah,

Gal Oren: You can you can you can you can decide to emit it eventually.

Samyak Jhaveri: that's what I was thinking. That's what I wanted to get your take on.

Gal Oren: Mhm.

Samyak Jhaveri: Um heat obviously took very less time because it's very simple relatively. And uh these are the run times,

Gal Oren: Mhm.

Samyak Jhaveri: build times and run times. Um, so yeah, moving on to um I think Errol should talk about the augmentation. So I'll move on to the parts that I know for sure.

Gal Oren: Mhm. So before before you'll move on uh I will give you a few few notes.

00:04:23

Samyak Jhaveri: Um,

Gal Oren: First of all for the paper you will need to have uh some technical some uh essential. You will need to have a new username. Open a new username named Powerbench in GitHub and make sure that everything is at that website because we need this to be an anonymous sub submission eventually and websites cannot be anonymous under a username. So if you will have a powerbench.github.io IO you will be able to put it

Samyak Jhaveri: Yeah.

Gal Oren: in the in the in the in the abstract of the paper and as such it will actually you know you you will have the benefit of the reader be able to read the website. So that's that's the first point. Second, um I think that in that website already we need to have um like one screen, one tab in which we demonstrate I don't know if you have it maybe I I missed it that we need to have if you remember the initial step that we did to come to this point which was taking all of the opensource uh codes out there and then to showcase which one connect to which one all of them things that I send in the in the in the initial mail remember.

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Gal Oren: So why we need that? Not because we will select all of those kernels, but because we will need to explain in the paper and in the website what is the um pool from which we are taking the uh kernels and why eventually we selected for parbench for now only those APIs and the subset of Hecbench and Rodinia. You will need to say something like although the pool from which I want to take codes is much larger in this and I already mapped what uh I already mapped the opensource relevant uh kernels that can be used for such um for such for such a benchmark. It is a lot of work. So I'm selecting a a representative subset and why it's representative because HB and Rodinia many use those in their evaluation. Anyway, there are enough kernels here. It's more than 100 kernels I think that you're using. The APIs that you are showcasing are the ones that came out as the most prominent ones, right? It's so academically speaking we need to have this layer at the beginning of the talk to explain why we came down to that

specific benchmark because one of the things that reviewer can tell immediately when they see it is why do I see that and not something else?

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Gal Oren: Why those kernels are being selected and not others? Why those specific APIs and not others? So the the claim will be I made a much bigger survey. I collected all of the benchmarks. I mapped their internal connections. I came to a conclusion. When I came to that conclusion, I decided practically to have a subset of that representative subset. And here it is. in which people can uh add their own kernels into it. Why? Because eventually we have the second stage of the augmentation. So anyone that want to add a a benchmark want to add a kernel to the benchmark can do that. And that's would be like a terrific thing to to to the to the framing of the of of the paper. Okay. So that's that's a that's one point. Um, what else do I I wanted to tell you at this step? Um, yeah, that's that's it for now. You can move on to the to the translations to the augmentation.

00:09:07

Samyak Jhaveri: Yeah.

Gal Oren: Sorry.

Samyak Jhaveri: So the augmentations from my end when I try to integrate them with the harness the augmentations are failing miserably. L3 and L4 are basically just failing but I have identified what bugs there are and I have made a report of it. So Erl is working on that and he's he's really prompt with helping me out and

Gal Oren: Thank you

Samyak Jhaveri: um um so yeah this is a mo this is a breakdown of

Gal Oren: everyone.

Samyak Jhaveri: everything. I I don't think I should get into the nitty-gritty right now for the uh for the meeting, but overall um there are immediate action items and optional forel. And um the last thing I'd like to talk about is um

Gal Oren: Mhm.

Samyak Jhaveri: okay, we went over

Erel Kaplan: I can I can add some uh some input on the augmentation now or later whatever you

Samyak Jhaveri: this

Gal Oren: Yes. Now,

Erel Kaplan: want.

Gal Oren: now we're now moving to talk about

Erel Kaplan: Yeah.

Gal Oren: augmentation.

Erel Kaplan: Okay. So, so the the thing about the augmentations is the code I got from Tom was as Samyak said it was meant for OpenMP.

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Erel Kaplan: So many many like as as you said like CUDA and OpenCL had some language related

Gal Oren: Hello.

Erel Kaplan: symbols and reserved words that it didn't dealt with it well and the code broke and another now I'm I am fixing it and I I think that tomorrow everything will be fixed. But the the main issue I see is that we we want at level four. What I'm testing is mainly level four because this is like the most aggressive augmentations and I think this is what what we'll we'll use the most to test the the actual understanding of the code of the LLM. So the the thing is that the the the augmentations it it needs to be conservative because we we want it to work with everything that that that that is it is given to us. So maybe we we we we will have a conservative h augmentations and that means that some names of some variables will not be be changed because the

Gal Oren: It's okay. It's okay. Just we will just need to to to say in the paper what was the scope to to uh uh the

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Erel Kaplan: Yeah.

Gal Oren: augmentation took took us. uh we don't we don't really need to do everything but we do need to say that um more augmentation could be done that also for a future work we can do more augmentations but for now we did only that amount of

Erel Kaplan: Yeah. So, so yeah,

Gal Oren: augmentations

Erel Kaplan: this is like a there's a tradeoff between how aggressive we we we want the

augmentation and how far we want to push the the because it is static augmentations. So, so there's a trade-off of how far we want to push it and how like

Gal Oren: Mhm.

Erel Kaplan: when when when we will give it the new code, we we want it not to break based on what the Yeah.

Gal Oren: Right.

Erel Kaplan: So,

Gal Oren: Right.

Erel Kaplan: so I'm working on that and based on the specs we we we already have, I'm I'm making sure it h augment everything well and not breaking anything.

Gal Oren: Are

Erel Kaplan: And I'm I'm I hope that when we in introduce new specs,

Gal Oren: you

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Erel Kaplan: it will still work well with them.

Gal Oren: okay? Great. Great.

Niranjan Hasabnis: So one thing I suggest about augmentation.

Gal Oren: I just I just said um shared in the in the chat that also something that is relevant for the uh the chat. I send it in the group chat. I will send it here as well that I'm reminding you that for the

Samyak Jhaveri: Okay.

Gal Oren: paper uh the baseline paper for us is the paraval paper. So I want you Samyak and to to to remember that that when we need to confront or to put a baseline for our work so virtually the paral's uh uh papers I think it's two papers now uh are the are the are the point of reference it will first of all it will make your life writing this much much easier because you already know what is expected from such a paper that's a first but a Second is that I think there are some essential changes because you will say that paravan took took it in a very naive way.

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Gal Oren: I would say they put the the bench correctly but they put it in a very naive way and what we are doing is something like a completely different evolution from that. So

you will need to keep in mind to read that paper on one hand but remember how we defer from that paper because paraveal eventually is not really useful for such cases although we did use this for the uh paracodics but eventually it's not augmented. Uh the kernels inside it are very few or not representative. There is like one big kernel that is actually important. uh there is no good explanation to why they chose those specifically

Samyak Jhaveri: Yeah.

Gal Oren: because it's the selection is not really good. We there there was a need to select others why those API why does it's like many questions

Samyak Jhaveri: Yeah.

Gal Oren: that that that needed to be framed differently. So I want you like to remember that that's like the the the baseline paper. Okay that's that's great. Um yeah, move on.

Samyak Jhaveri: I um I have a question for you guys and I think Nanjen can help me understand better.

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Gal Oren: some

Samyak Jhaveri: Um see for the GPU APIs um as I said in the chat like a little late in the night is it's not just kernel time we're looking at we're looking at how well an LLM translates the entire code. For example, if you have two models, one model writes really fast kernels but is terrible at data management.

Gal Oren: Right.

Samyak Jhaveri: It's going to be a very slow program as a whole because data management is the major latency introducing um aspect of parallel code on when using GPUs. Another model writes a so or a decent GPU kernel.

Gal Oren: Right.

Samyak Jhaveri: So kernel time is a little more but then it it is really good at managing the data and

Gal Oren: Mhm.

Samyak Jhaveri: that is very situational based on the input size based on the algorithm you're using based on what combination of kernels you're using. So my my theory not my theory but my um proposal

Gal Oren: Right.

Samyak Jhaveri: is that we do report kernel times but we also should report uh overall program time because in an ideal world we don't run individual kernels we

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Gal Oren: Yes. Yes. Do it.

Samyak Jhaveri: run okay sorry Yeah,

Gal Oren: We have been convinced. You can do it. Yes. It's it's it's okay. No. No.

Samyak Jhaveri: I

Gal Oren: It's it's it's it's great. It's it's it's a good

Niranjan Hasabnis: No but I think there is a subtle difference right what do you mean by end to

Gal Oren: point,

Niranjan Hasabnis: end time because when you in Linux you when you measure time right it basically tells

Gal Oren: right?

Niranjan Hasabnis: you walk clock time and then there is also CPU time so CPU time is when the process is simply executing on a CPU right but walk time also includes the wait time so if

Gal Oren: Yes.

Niranjan Hasabnis: process is blocked let's say on a network connection or something or let's say fetching data from disk that is still like a wait time right so we don't want to measure that time so system CPU and user CPU time

Samyak Jhaveri: That's

Niranjan Hasabnis: is I think reasonable to measure and I think at least in terms of like GPU kernels I have seen two metrics one is the kernel time itself and the other one is kernel plus the host time so host memory management and all that thing so transferring data between host and a device and copying back from device to host so that is the end to end time for a for a GPU kernel

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Gal Oren: Mhm.

Niranjan Hasabnis: Yes.

Samyak Jhaveri: So that's what I wanted your help

Niranjan Hasabnis: Yeah.

Samyak Jhaveri: with.

Niranjan Hasabnis: Sure. I think we can come up with a way to measure that.

Gal Oren: Huh?

Niranjan Hasabnis: I think there is way

Erel Kaplan: Also also as I as as as as I as I wrote we can use the the metric we used in a paracordex paper the GPU time matrix that includes the uh kernel execution time plus the memory transfer

Gal Oren: Yeah,

Erel Kaplan: time.

Gal Oren: we we already had this discussion many times. So Ariel and John, we we can we can do what we did and add some additional. It's okay.

Samyak Jhaveri: Okay. So, I'll add those I'll add those metrics from Paracordex paper and I'll put them here as well. Um,

Gal Oren: Mhm.

Samyak Jhaveri: lastly, I built like a little dashboard for everyone to, you know, help be on track.

Gal Oren: Mhm.

Samyak Jhaveri: um considering April 8th as our deadline and um we have 22 tasks and so on and uh yeah so just I think this is a quite

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Gal Oren: So what let's so now after we talked about the uh

Samyak Jhaveri: useful

Gal Oren: curation part the um um augmentation part and now and the and and now we are moving to the to

Samyak Jhaveri: Yeah.

Gal Oren: yeah now we're moving to the to the translation evaluation. Let's talk about that. So I gave you the 4.1. I think that's a good model to check because it's virtually the last one which is not reasoning. Um but but please everyone what what do you think except I mean given the time that we have except GPT4.1 what are the other models that we need to put in the paper to make sure it will be will be accepted.

Samyak Jhaveri: A typical criticism uh that most people get is we should do more open source at least one open source model if our narrative is we want it want to open source this we should have at least one open source model

Gal Oren: Okay.

Samyak Jhaveri: evaluated I can do

Gal Oren: You can do you can do llama.

Samyak Jhaveri: llama

Gal Oren: Okay.

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Gal Oren: What else?

Tomer Bitan: It might also be wise to get like some agentish model like maybe a CEX or

Samyak Jhaveri: Um,

Tomer Bitan: something.

Gal Oren: No, I think that's that's a good point. I think that that given the time and the frame we should not I think I think and the task I think that for this paper we will not do we will not perform models that that um we will need to like contemplate a lot about the the results. I think that the direct models are are the best ones for this because I will say not not I mean not even because of the time I mean if we had enough time I would put us in two two different categories because eventually what we are testing is the ability of of the model itself to we want to see what the model itself not the reasoning part of it what the model and without like contemplating like based on sources and in the internet and so forth what the model itself knows about this those benchmarks So and and that's why the augmentation part is also so so important.

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Gal Oren: We want to see different difference in results in the augmentation part. Right? If we will give it now if we will test it over the reasoning. So a a model can quickly reason about the code that is actually irrelevant the the the the changes of the of the variables and and and understand that this is actually a code that it can match to a code that is already known. And by that we will have a huge problem with the with the evaluation part. It's not it's not that I don't want to evaluate this on agents and and and that kind of reasoning models. It's just for that specific point in time and for the completeness of the claim of the paper. I think it will make it much much more hard to to take that into account now. So I think non-reasoning or non-agentic of course uh uh models for now would be would be enough. So Llama 70 GPT 4.1 uh name others that you want to

Tomer Bitan: And he told her

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Gal Oren: test.

Niranjan Hasabnis: I think we don't need to go after all the models like that.

Samyak Jhaveri: most

Niranjan Hasabnis: There's no end to it, right? So, we should just pick maybe two and then stick with that. Like,

Samyak Jhaveri: I agree.

Niranjan Hasabnis: open AI should be okay. Um, but yeah, I'm not I wouldn't say that we want all combinations too many.

Samyak Jhaveri: So

Le Chen: Um so yeah um there there there are multiple there are different

Gal Oren: Mhm.

Le Chen: uh leaderboard for coding capabilities. I would like to check those and pick top um top ones.

Samyak Jhaveri: Should

Le Chen: That would be I mean we that's that would be reasonable choice. We can we could also argue that in the paper like we picked from some leaderboard um when we're selecting the modals. I I saw um I saw um Q1 is there for for open source models. We I saw um Gemini sorry not Gemini Q1 was there and Mistral was there. Um yeah you can you can check them just uh check one leaderboard or whatever.

00:22:43

Le Chen: So that could also um re reasonable our make our selection reasonable.

Gal Oren: Mhm. Yes.

Samyak Jhaveri: should we include a code model like a a code specific model or just stick with general purpose about

Le Chen: Um we we don't have to we don't have to do that.

Gal Oren: model.

Le Chen: Uh we don't have to like that's our not focus. We just focus on the performance of the models on general code tasks and their their their ranking on a leaderboard. Some code models they have a they don't have a good performance compared to general models. That's fine. We can just pick General Motors.

Gal Oren: in

Niranjan Hasabnis: Maybe another option would be to compare against uh model specific to parallel API

Gal Oren: green.

Niranjan Hasabnis: transition from the prior work. I mean we need to have something on that end also.

Gal Oren: Come

Niranjan Hasabnis: I mean we need to also compare with the prior work right.

Gal Oren: again.

Niranjan Hasabnis: So we need to have pick some model which is designed for parallel gen.

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Gal Oren: Ah yes. What what what which model they used in the in the power evolve? Did they use some what what was this that they used? Let's see.

Le Chen: No, I don't think power using any purpose

Gal Oren: Ah yeah. Yeah. I I I see it now. They used Gemini GPT4 lama 3.3

Le Chen: word.

Gal Oren: Quen and that's it. Maybe maybe it would be good to match their own I mean even if it's like a little bit more advanced but use use their own uh models that they test. That's that's the one. Let me share it. I don't know how to do this

Le Chen: Um,

Gal Oren: here.

Le Chen: and goal I I I totally agree with you that we we should not worry about the agents or agentic modals, but uh I'm a little bit concerned about the uh rhythming models.

Gal Oren: Mhm. Do you see my screen?

Le Chen: Yeah.

Gal Oren: So, those are the ones, those five.

Samyak Jhaveri: I

Gal Oren: Mhm.

00:24:55

Le Chen: is that the party wallpaper.

Gal Oren: That's the power of all. Yeah, I showed it here in the chat.

Le Chen: Yeah, it's I think it's been a it's been a while since uh their paper.

Gal Oren: Of

Samyak Jhaveri: Okay.

Le Chen: uh one one big yeah one big difference is uh is there has been multiple rhythming models

Gal Oren: course.

Le Chen: even for uh the same model you can pick if you want to turn on the reasoning or you you don't want it on so that's something I think uh we need to discuss do we need to include it or we just keep everything without a reasoning

Gal Oren: Ah, okay. That's also good. Yes, that's that's also a good part. just just we need just need to have the explanation about this. But yes, it's it's it's a good suggestion.

Le Chen: Yeah.

Gal Oren: Yes.

Le Chen: Um Sam I I think we can start from without a reasoning and if we our time or the or the result whatever we limitation we have um we can we can argue

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Samyak Jhaveri: Got

Le Chen: that uh for fair comparison we turn off the the reasoning for all the all the models and uh if we have time and there are some interesting result we can find we will include them later but for right now just don't do reasoning. Cool.

Samyak Jhaveri: it.

Le Chen: Yeah. Thank you.

Niranjan Hasabnis: So, can you just tell me the story?

Gal Oren: Uh everyone this is Lan. Lan hello to everyone. This is our group. Luran is a student at the Techneon uh that finished masters and he's helping Tor in his effort. So in a while they will explain uh about that but I want to first finish with Samyak. Um okay great. So Samyak anything else that is open uh question for you before you'll move on to the last stretch because from now on that's the last stretch. Basically our next meeting uh will be to revise the draft of the paper.

Samyak Jhaveri: Okay.

Gal Oren: What what other open question do you have for the group before

Samyak Jhaveri: Yeah.

00:26:54

Samyak Jhaveri: Yeah. I'm reading my notes.

Gal Oren: that?

Samyak Jhaveri: So, we have two suites for the paper, Rodinia and Hickbench.

Gal Oren: Yes.

Samyak Jhaveri: Should we attempt a third one?

Gal Oren: Yes. It's it's it's not it's even it's even not all of that.

Samyak Jhaveri: Yes.

Gal Oren: You will you will omit some part of Gordinia, some part of Headbench. But but as I told you, remember the paper should start from the curation part and and as as with the email that I sent you back then and and really

Samyak Jhaveri: Yeah.

Gal Oren: explain first of all what should be benchmarked. The paper should start from that fundamental question because others think about it.

Samyak Jhaveri: Yeah.

Gal Oren: All of the other papers that we are talking against them just select um uh that selectively took some some benchmarks out there and use those right instead of saying look I systematically reviewed the the web. I I understood which APIs and which benchmarks out there are relevant. Then I took representative subset that's your claim now. So you will start by saying that you start by saying I have all of those benchmarks that I found given a systematic open source that all of those somehow relevant.

00:28:10

Gal Oren: Then I looked over all of the APIs. I managed to focus on specific core of APIs that seems to be very relevant. I'm taking those specifically and I take subset of that and that will allow others to expand that in the future. But but that should be like crystal clear the the intro and and rationale of the

Samyak Jhaveri: Yeah. uh so essentially I will take inspiration from uh the work that you did at the initial what you showed

Gal Oren: paper

Samyak Jhaveri: me like the graph and uh honestly that in itself is such a nice paper like it's a nice five-page paper uh with that

Gal Oren: but it will but it will not be enough for supercomputing.

Samyak Jhaveri: analysis yeah that's true that is true yeah

Gal Oren: Supercomputing won't all the way through.

Samyak Jhaveri: so okay uh I think that u that world building is very important so I'll just make sure to add that

Gal Oren: And you have and you have enough space. Supercomputing is a is it's a super nice conference. You have double column 10 pages and I think you can have as much as appendices that you want.

00:29:08

Gal Oren: So virtually it means that you can put all of the results that you need in that paper. It's it's it's really like one of those venues that still you can you can put enough info in that paper.

Samyak Jhaveri: Got

Niranjan Hasabnis: So are we saying that we are building like a super benchmark for for par computing? I mean I'm trying to understand like how we are positioning the paper with respect to like paral and other benchmarks right but I think what what you said based on that then we should have a similar

Gal Oren: Mhm.

Niranjan Hasabnis: analysis for paral and other benchmarks so that we can compare like how many APIs they using or like how much is that with respect to like a kuda or something right?

Gal Oren: Yeah. the parallel cases. The parallel case is so is so easy. They're used in I don't know like four kernels or something like that and three of those are like super small. It's it's almost about a metric paper than an actual benchmark paper. Parval talks about the evaluation not actually suggesting a benchmark.

00:30:09

Gal Oren: That's there is no uh please tell me if I'm wrong about this. There is no specific benchmark paper for that code translation task. Am I right?

Le Chen: Um I agree with you especially there are multiple APIs to APIs so there's no no paper talking about

Niranjan Hasabnis: No,

Gal Oren: Mhm.

Niranjan Hasabnis: but there is I think par itself is a benchmark,

Le Chen: it

Niranjana Hasabnis: right? So they they you can basically take their code and if you run it against LLM, it's going to generate input kernels for different APIs and then evaluate

Gal Oren: No, it's a it's a it's a Nerjan. It's just Nanoxaur, Microaur, Microaur, and EXS bench. That that's it. It's it's um it's it's almost irrelevant at at that part. It's it's mostly talking about how to benchmark and not that those benchmarks are

Tomer Bitan: And one of them doesn't work.

Gal Oren: good. The excess bench is the actual app there. The rest is is not even

Niranjana Hasabnis: Okay.

Gal Oren: enough

Niranjana Hasabnis: So we we I mean essentially I think we are saying that we are basically curating a benchmark between parallel APIs, right?

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Niranjana Hasabnis: And then that's I mean that's the main contribution of the paper.

Gal Oren: right. We we yes we can also add Samyak I'll told you that also you can we can also add to the paper that there is like another work um uh also from uh Battel's uh group uh I forgot the name of that work Njan you obviously remember this the work that they try to generate kernels using um using AI like two years ago there was a work about this right They devised small kernels by prompting an N an AI system to give them like a collection of kernels to

Samyak Jhaveri: Yeah,

Gal Oren: test.

Samyak Jhaveri: I remember that paper. Yeah, they they got some model to generate CUDA uh and then they did like

Gal Oren: Yeah.

Samyak Jhaveri: um multiple CUDA generations. They tested them, profiled them and found out which ones were the best,

Gal Oren: Mhm.

Samyak Jhaveri: combined them and then used that as an RL uh loop to train a smaller.

Gal Oren: Mhm.

Samyak Jhaveri: That was their whole complicated set setup um couple

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Gal Oren: Yes. And it's not it's not only comp it's not only it's not only complicated.

Samyak Jhaveri: years ago. Yeah.

Gal Oren: It's it's it's it's completely different. It it takes the baseline as let's create from the AI model a code instead of let's curate

Samyak Jhaveri: Yeah.

Gal Oren: human code and try to see how we benchmark against it. So I'm trying to to find this paper as well.

Niranjana Hasabnis: I recall even parl also has the augmentation strategy right.

Gal Oren: Yes.

Niranjana Hasabnis: So they um I mean I I I recall the point that they say that because the code is out there in the control we augment the code so that we can evaluate better.

Gal Oren: Okay. So just just remember Samyak to have to have a section in the paper uh for previous pre of of like a literature review previous work section. It it is important to have that. I mean after the intro uh the resource questions and the contribution section you should have a a previous work section. This paper needs that.

00:33:34

Gal Oren: You need to explain specifically how this paper differs from previous papers that used or or devised or suggested some kind of some types of of benchmarking for even relevant or or similar uh um problems. Okay. Put put this as as something that you must have in the paper.

Niranjana Hasabnis: And I think one thing I will say is that sorry for the background noise. My son is shouting in the background. But I think one thing I will say is that if you want to really compare let's say our benchmark against existing benchmark then we should I mean one takeaway that would be interesting would be to see whether LLM that perform better on their benchmark perform worse on our benchmark or vice versa. Right? But that tells that tells the fact that the benchmark itself is not good with respect to evaluating a right.

Gal Oren: It's interesting.

Niranjana Hasabnis: I think in that sense maybe we should compare against the same model that they use in their paper right so that we can then see the performance of the alert model over bit and

00:34:39

Gal Oren: Uh Samya, can you can you use excessbench as well in in your uh in your benchmark because that's like the main kernel that they used in in Paral.

Samyak Jhaveri: X's bench.

Gal Oren: That's like that's the main one. The rest are not really true.

Samyak Jhaveri: Okay.

Niranjan Hasabnis: not.

Gal Oren: And then you can you will be able to say look I I can have like decent comparison to previous state-of-the-art something.

Niranjan Hasabnis: So I would say instead of I think going to yet another benchmark to add to far maybe we should just create the outline of the paper and that will give us a sense of like where or how we can fine-tune it or position it with respect to existing work right I think just adding yet another benchmark to our u itself may not be good use of the time so far

Samyak Jhaveri: Yeah, I agree.

Gal Oren: Mhm.

Samyak Jhaveri: I'm noting it down. Um, so first we need to create an outline of our paper and see where the gaps are or where we can um improve upon existing work and how our work is different.

00:35:51

Samyak Jhaveri: We should emphasize on the related work section and the previous work section in benchmarking LLM for parallel code generation because uh it is very important that we know where we stand and

Gal Oren: Mhm.

Samyak Jhaveri: how we have improved over existing work.

Gal Oren: H

Samyak Jhaveri: uh Excusbench is a very strong candidate for the third benchmark to use and um moreover we have to explain how we got here to the point that why did we select these benchmarks these APIs and what kind of survey your study did we do that justifies the use of these um and how we are

Gal Oren: There's

Samyak Jhaveri: focusing on the ones that are relevant. Rodinia is huge. Hebench is huge. But we got to focus on uh the important kernels and important translations

Gal Oren: enough

Samyak Jhaveri: because of time and compute resources and the breadth instead of depth. the breadth of benchmarks that we look into versus the depth of benchmarks that we look into that are enough to uh establish our narrative of open-source API to API benchmark that is relevant to all uh LLMs and agentic systems in the future and why we are only doing for non-reasoning models for this our paper to limit the scope and to identify

00:37:05

Gal Oren: Mhm.

Samyak Jhaveri: its knowledge inherent knowledge and that's where the augmentation explanation comes in that we don't want it to work based on memory. We wanted to based on uh we wanted it to work on based on its intelligence and understanding and that is why we are doing the augmentation part. As the augmentation gets more aggressive the pass rate reduces which is expected and

Gal Oren: Mhm.

Samyak Jhaveri: um we have to prove it for two or three kinds of LLMs. one is GPD 4.1 the late the last non-reasoning model uh llama 70B I might have to get help from model uh on hosting that because

Gal Oren: It is my

Samyak Jhaveri: my uh setup cannot host that and third is find uh top five coding models that are general purpose from one of the leaderboards

Gal Oren: Mhm.

Samyak Jhaveri: as lechen said le said and use one of those models that are practical for our timeline and um compute constraint.

Gal Oren: Perfect.

Samyak Jhaveri: Got it.

Gal Oren: Perfect.

Niranjan Hasabnis: Yeah,

Gal Oren: Okay. So,

Niranjan Hasabnis: maybe just pay attention also to the fact that like different say some of the models that

00:38:10

Gal Oren: yes.

Niranjan Hasabnis: existing benchmarks have used right like paral for the sake of comparison and uh I think to to the whole group we should also find out all the relevant benchmarks

Gal Oren: Yeah.

Niranjan Hasabnis: here we need to refer in the paper right I'm not familiar with maybe all of them but in case you guys are then feel free to share

Gal Oren: Perfect. Perfect. Okay, that's great. So, Samyak, great work. Um I want to move on to Tomer for the next 15 minutes uh for the other project. So Tomer I remind everyone Tomer is working for a different venue. He's working for the new venue. So we have more time for that. So that's why we devised only 15 minutes now. Um uh but after the submission in 3 weeks or le some three weeks we will talk mo mostly about that. So Tomer give us an update on your uh

Tomer Bitan: Yeah. Uh,

Gal Oren: status.

Tomer Bitan: so I'm so like, okay, I I expected to be uh further along at this point.

00:39:25

Tomer Bitan: I I was hoping to have first results at this point, but I was not expecting how long it takes to build the the the data set. So I have like 220 kernels already run,

Gal Oren: move

Tomer Bitan: but each one takes from 10 to to 20 minutes to run. And so I I was trying to like supplement this with a because I realized it it might be a small data set as is. So I'm trying to supplement it with a a secondary data set.

Gal Oren: to I remind you that you need to give context from the beginning to what you are doing.

Tomer Bitan: So okay. Yeah. Yeah. Yeah. Sorry. Uh so I'm I'm building the the data set to train uh the the guidance model that's sitting on top of the main model and it's trying to look at the latent vectors and assess how good the reasoning steps are. So for that I I was I'm working on making a data set of taking Hecbench running it through a

Gal Oren: Right.

00:40:15

Tomer Bitan: llama 70 billion and saving the reasoning step uh vectors uh to the side and giving each one a score.

Gal Oren: Mhm.

Tomer Bitan: So uh a few weeks ago with and we were working on all the infrastructure for that and we got that working and it's up and running.

Gal Oren: Ah, that's good.

Tomer Bitan: The problem is that it takes each Yeah.

Gal Oren: Okay, great.

Tomer Bitan: So so that's working and Lean finished her part and we finished his part.

Gal Oren: Yes.

Tomer Bitan: Um, and now I got Lean back uh for like starting uh a different sub project with his whole thing for like the next part. Um, and also Lan is joining at me um to help me with uh all the a lot of work that we still have left. So um yeah, so like I was saying earlier, the challenge right now is that it's taking a long time to generate each sample that I need in order to train the PRM. So I have a little mini data set that I made with like 10 kernels and I figured out that like my my code is running correctly, but I still don't have a trained like process guidance model.

00:41:20

Tomer Bitan: Um which I was expecting to have at this point. And uh yeah, so I'm slightly conflicted

Gal Oren: But basically you're saying but basically you're saying that it's already running.

Tomer Bitan: between

Gal Oren: So now it's a matter of time. So virtually over the weekend you will have the results.

Tomer Bitan: I mean so creating the the the the data set is running and then I have to make to train the the little the the baby model the the guidance model which will be like probably a day not no not more than that.

Gal Oren: Huh?

Tomer Bitan: Um the problem is that I don't know if it's it's a bit hard to estimate I if the hebench will be enough uh kernels does it have enough data to train the PRM. So I'm trying to to add a second that data set that is easier to to make.

Gal Oren: Mhm.

Tomer Bitan: So I went to the original data set that all the other process guidance

models are working with which is math-based. Um so I uh in the last day and a half I added like I have on

00:42:15

Gal Oren: Mhm.

Tomer Bitan: modal running uh thanks to Nanjan uh one that's generating a similar type of data set which is llama 70 billion asking questions saving the the vectors but it's running uh on the math data set. So,

Gal Oren: Mhm.

Tomer Bitan: I'll I'm hoping to supplement the how good is this thought process with at first like pre-training on math questions and then on the code questions because I'm worried that a it'll take too much time by the time I get my code data set

Gal Oren: Mhm.

Tomer Bitan: ready and b it'll not be enough uh kernels. So cuz at this rate I think it'll take another more

Gal Oren: Mhm.

Tomer Bitan: than 3 days like maybe even five or six six days it depending how much how fast I can get Lan to help me with joining his machines and how much I'm wanting to spend on adding uh Nanjen's model machines because right now I have one run on my machine and occasionally I get enough resources from the Newton gods to to give me like the testing on another computer and

00:43:27

Gal Oren: Can can we I mean nan can we can we find a way to give him just one more

Tomer Bitan: then

Gal Oren: server from the the cloud that you're working with and and by that to to help him out. I think that that will allow you to parallel the work right you are working with one server

Tomer Bitan: yeah right now I have like uh I

Gal Oren: now

Tomer Bitan: just added the the modal and I tried to run the the math data set on the modal because it's a bit smaller. So, I'm expecting it to to be cheaper like to to save on time.

Gal Oren: and another another would would would one more server will be

Tomer Bitan: Um,

Gal Oren: will be good for Can we help him with one more

Tomer Bitan: yeah, it would be helpful. Yeah. Uh,

Niranjan Hasabnis: But I think you can in in mono you can actually have more like multiple GPU

Gal Oren: server?

Niranjan Hasabnis: instances,

Tomer Bitan: yeah. Yeah.

Niranjan Hasabnis: right?

Tomer Bitan: like um more of a permissions thing like how liberal is it okay to to add another run if if I need it for the for the data set or are we very tight

00:44:31

Niranjan Hasabnis: Yeah,

Tomer Bitan: about adding runs?

Niranjan Hasabnis: not really tight I would say but I think uh another parallel one should be enough.

Tomer Bitan: Uh okay.

Niranjan Hasabnis: Um I mean I can check with uh Alex from our code medicine who is

Tomer Bitan: Okay.

Niranjan Hasabnis: into provisioning part but I yeah in model you can have multiple

Tomer Bitan: Okay. Thank you.

Gal Oren: Thank you John.

Niranjan Hasabnis: GPU instances I do

Tomer Bitan: Yeah. Yeah. Okay. So,

Gal Oren: Right.

Tomer Bitan: I'll have to make like a small modification to my code for that to work right now because it's only for the math, but yeah. Okay. Uh yeah, perfect. That that'll help speed things up. Um yeah.

Niranjan Hasabnis: And regarding

Gal Oren: One more thing that I want to ask you to at the meantime um given given

Tomer Bitan: Mhm. Yeah.

Gal Oren: that things are running at some times as I asked Samyak as well I want you to

Tomer Bitan: Yeah.

Gal Oren: open uh that that's helps a lot for clarifying for the rest of the group and for

me and for you uh open an overleaf project and new overleaf project

00:45:21

Tomer Bitan: Okay.

Gal Oren: already and the things that you already know about this work.

Tomer Bitan: Mhm.

Gal Oren: Try to already write them.

Tomer Bitan: Mhm.

Gal Oren: So have a title,

Tomer Bitan: Okay.

Gal Oren: have a fake abstract, have an intro, put the research question, put the methodology even though that you still don't have results or you are not fully sure how the training and testing going to going to come out.

Tomer Bitan: Mhm.

Gal Oren: I do want you to already given the fact that you you have that now to have a systematic view on that.

Tomer Bitan: Yeah.

Gal Oren: So it will help the rest of us to to more to be

Tomer Bitan: Mhm.

Gal Oren: able to be more able to help you.

Tomer Bitan: Yeah.

Gal Oren: Okay?

Tomer Bitan: Okay. Okay. Yeah. So,

Gal Oren: And it will help you to collect the things and put those uh linearly in

Tomer Bitan: okay.

Gal Oren: in the paper and that will help you you know to move to move on uh in in in a

Tomer Bitan: Yeah.

00:46:21

Tomer Bitan: Yeah. Yeah.

Gal Oren: pace that's that is

Tomer Bitan: Mhm. Yeah. Now that now that Lan is joining me,

Gal Oren: trackable.

Tomer Bitan: I was planning on doing Yeah. Basically, you you said it better than I was thinking it. Like the the plan of what exactly I want to have in the final paper and then I can send

Gal Oren: Perfect.

Tomer Bitan: Lan ahead to do some of that stuff while we're running uh in parallels. Um,

Gal Oren: Perfect.

Tomer Bitan: so I I was I I had a question for you guys if if we have more time like So I was scared that so I have six weeks until the the new deadline because it's uh on

Gal Oren: Yes.

Tomer Bitan: the sixth or or 9th I think of uh May. Um, and I was thinking about 3 weeks of paper writing with like supplementing a few maybe tests somewhere and that gives me three weeks to basically finish the tests like everything I the experiments I want to run and I'm scared.

Gal Oren: makes sense.

00:47:20

Tomer Bitan: So, I'm a bit scared I won't make it. And I'm thinking,

Gal Oren: Yes,

Tomer Bitan: is there a pivot? Like something you think that I've been talking about and maybe we can like slightly pivot it to drop it, make it simpler or uh

Gal Oren: that's exactly that's exactly why I wanted you to write to to immediately start writing the paper so we

Tomer Bitan: not Okay.

Gal Oren: will be able to feedback exactly on that in two weeks from now you will get

Tomer Bitan: Okay. Okay.

Niranjan Hasabnis: I think one thing I

Gal Oren: a specific even less than that if you will send the uh over leaf in I don't know few days we will be able to reply uh over email and give you

Tomer Bitan: Mhm. Okay. Okay.

Gal Oren: like exactly what we think about where what you're currently doing and by that we'll be able to

Tomer Bitan: Okay.

Gal Oren: either to guide more or tell you drop that put more emphasis on on the other uh directions and so

Tomer Bitan: Yeah. Okay.

00:48:09

Tomer Bitan: Got you.

Gal Oren: forth.

Tomer Bitan: Thanks. So,

Niranjan Hasabnis: I think one thing I can also say to that you were wor worried about whether he bench is enough for

Tomer Bitan: yeah.

Niranjan Hasabnis: your experiment right so maybe the way to approach that would be to reduce the model size make it smaller so that you don't even like headbench is bigger for the model so you can actually take a subset of the

Tomer Bitan: Good.

Niranjan Hasabnis: hbench and experiment with whether the idea actually works on a smaller model and then move to the bigger model so instead of giving going to like 70 billion start off with something like 8 billion model.

Gal Oren: Good.

Niranjan Hasabnis: Take a subset of headbands and see whether it works. If it doesn't work, then there's no point in going to 30 billion anyway.

Tomer Bitan: Yeah, the problem with the 8 billion model is that we saw like earlier in the UNIPO paper that an 8 billion model is basically useless on hikebench like it was doing nothing.

Gal Oren: All right.

Tomer Bitan: So, uh, from what we we thought in the beginning, the reason we wanted to go for like I I I I see what you're saying and I'm thinking maybe I'll try like if you think it's a smart idea, maybe I can try the Quen 30 billion or something, but I remember when we tried and now when I'm running the 8 billion model.

00:49:11

Tomer Bitan: So, I was doing a lot of like mini tests with the 7 billion and stuff. Uh,

Gal Oren: Mhm.

Tomer Bitan: that was like it was basically like not even finished code. It it was like snippets that don't even end. So, I'm a bit scared that like you'll improve from nothing to a bit more than nothing, but it's still nothing. Like I I won't even get like the full compiled

code to to see the number go up with the 8 billion.

Gal Oren: I was wondering

Niranjan Hasabnis: I was wondering whether there is any kind of relationship between the smaller model and the bigger model like size

Tomer Bitan: Uh,

Niranjan Hasabnis: wise and can help us in terms of training processes.

Tomer Bitan: sorry, I I can hear the last sentence.

Niranjan Hasabnis: I was thinking whether there is a kind of relationship between the smaller model and the bigger model. So let's say 2 billion, 4 billion, 4 billion, 70 billion, I don't know. And then based on that,

Gal Oren: I know

Niranjan Hasabnis: if you do the smaller model that can help us train faster, right?

00:49:58

Niranjan Hasabnis: And do experiment faster.

Tomer Bitan: So, so you're saying no. Okay. How do I measure uh the like say the current parameters that I know how to measure on hebench is compilation. If it passes compilation then the validation rate and then uh I have if it passes compilation then lean added variable comparison similar to howl did in the in the publiclex paper and I have uh um embedding similarities.

Gal Oren: I'm going to go to

Tomer Bitan: So the only one that works without compilation is embedding similarities.

So if my 8 billion model uh performs so bad that it doesn't compile ever, then the only measure for how good the code is is embedding similarities, which I'm a bit hesitant to use as like a full metric for a paper, right? Because it it sounds like not too precise.

Maybe maybe I should try Okay. Okay, I see what you're saying. Maybe I should try to like take it down a notch from the 30 from the 70 billion to the 30 billion. See if it gives me good code.

00:51:04

Tomer Bitan: If it does go down another notch to the 14 billion, see if it gives me quality code. Then it go down a notch.

Gal Oren: Go going going down is better then uh under then con yeah no then

Tomer Bitan: It's going up.

Gal Oren: constantly think that something is wrong with your method by using a small model and then need to go up I mean it's it's especially when you test a new a new idea that is

Tomer Bitan: Yeah. Okay.

Gal Oren: risky so the model itself might be you know problematic to be to begin with and you think that

Tomer Bitan: Yeah.

Gal Oren: you have a problem but it's actually not

Tomer Bitan: Yeah. Okay. Yeah. So, last thing I'm worried about this aspect is uh I do I didn't have a test yet to see if saving a vector like learning over a vector of um say a 70 billion model does it translate to when I'm inferencing on a 30 billion model. So essentially does does this mean that I need to remake all the kernels that I previously made which might not be

00:51:56

Gal Oren: H.

Tomer Bitan: horrible if it's significantly faster but if it is

Gal Oren: Start without doing so. Start without doing so.

Tomer Bitan: then

Gal Oren: Test that and then move on. It's very it's a it's a experimental question.

Tomer Bitan: start with

Gal Oren: Start without doing so.

Tomer Bitan: sorry

Gal Oren: Use those as as they are and then if it's not working for you

Tomer Bitan: okay okay no but but it means that I need

Gal Oren: escalate.

Tomer Bitan: to rerun all the that I already did. No.

Gal Oren: Yes. Do reevaluation.

Tomer Bitan: Okay.

Gal Oren: Yes.

Tomer Bitan: No. Uh, okay.

Gal Oren: before we Okay, so thank you Tomer. Before we will conclude,

Tomer Bitan: Okay.

Gal Oren: I have like two more minutes. So I just want to uh add Ariel,

Tomer Bitan: Okay.

Gal Oren: we have um the ACL paper acceptance notification on April 4th. That's exactly the date on which the uh supercomput submission ends. So you need to be ready with the supercomputing template in case the ACL paper is not accepted.

00:53:09

Gal Oren: Okay? Because you want to immediately move it there. I just like reminding you that that's something that you need to have in mind. It's mostly like template and you know like small modifications but just just make sure that you are you are uh on on on par with that. Um I think that the abstract submission is April 1st even. So also put it in uh in the calendar or something. Um okay I we're coming to the end of the meeting because I have to move also to another meeting. Thank you very much Tom for the the presentation and the work. Uh I hope that in in in a week we will see the initial paper and the uh in two weeks some results.

Niranjan Hasabnis: Just a thing I wanted to mention. I think Roman I think we are a bit confused about some of the things. So if you have any questions feel free to we can have a meeting separately for discuss if you need some

Tomer Bitan: Oh

Niranjan Hasabnis: inputs.

00:54:24

Tomer Bitan: yeah.

Niranjan Hasabnis: Yeah we can talk tomorrow or depending on like whatever you have.

Tomer Bitan: Okay. Yeah. Thank you.

Niranjan Hasabnis: Okay. Yep.

Tomer Bitan: Um so so let if you have a second now I have a quick

Niranjan Hasabnis: Thanks Le.

Tomer Bitan: question.

Niranjan Hasabnis: Um I don't have much time but I can talk maybe a minute or so.

Tomer Bitan: Okay. So do you think um so I I thought maybe I'm I'm a bit worried about rerunning everything in case it's not significantly faster. So do you have a strong intuition whether it's like it's going to be okay kind of thing like downgrading the model size and rerunning everything or should I be cautious about it and like start see how fast it goes. Is it significantly faster? If not give up and stay with the big model. Um, or do you or or

do you think it's like for sure good enough and and will

Niranjan Hasabnis: Uh I mean I don't have a like strong sense of exactly how things are

00:55:15

Tomer Bitan: run?

Niranjan Hasabnis: working currently. I mean at least based on u yeah I mean what you mentioned I know like a strong intuition like what's what's happening maybe if you can describe me

Tomer Bitan: Oh, okay.

Niranjan Hasabnis: the your findings so far over email maybe I can talk about

Tomer Bitan: Okay.

Niranjan Hasabnis: that maybe I can talk about that or maybe yeah email would be better

Tomer Bitan: Yeah. Okay. Okay.

Niranjan Hasabnis: in case like I need to understand like how you are experimenting and what you have found so far right and maybe based on that we can decide like what would they mind? Yeah. So before I understand that you don't want to repeat the experiment. So in that sense if you can describe me like your finding so far maybe we can take a discuss the next step.

Tomer Bitan: Okay. Okay. Yeah. Okay. I'll start by trying and I'll and I'll message you when I when I have like something concrete.

Niranjan Hasabnis: Yeah. Okay. Right.

00:56:15

Niranjan Hasabnis: So I mean the whole idea I was saying was instead of starting with a bigger model and then experimenting with it because it takes

Tomer Bitan: Okay.

Niranjan Hasabnis: a long time to train and get the results. Right. Maybe smaller model will be a lot easier faster to get the training and finetuning done. That's the whole idea.

Tomer Bitan: Yeah.

Niranjan Hasabnis: But I I I do not fully understand like what you are finding so far and how how does

Tomer Bitan: Okay. Okay.

Niranjan Hasabnis: that matter for the paper or

Tomer Bitan: Okay. I got you. Okay. I'll Okay. Yeah. I'll I'll do the experiments. I'll think about it and I'll I'll message you a clear

Niranjan Hasabnis: If you can summarize this so far,

Tomer Bitan: question.

Niranjan Hasabnis: maybe we can can take the next step.

Tomer Bitan: Okay.

Niranjan Hasabnis: Yeah. What we have found so far.

Tomer Bitan: Yeah. Okay, because it's like I don't have any results so far.

Niranjan Hasabnis: Yes.

Tomer Bitan: It's only like um rushing to create the data set that is the problem.

00:57:04

Tomer Bitan: So, so yeah. But uh yeah. Okay. Okay.

Niranjan Hasabnis: Uh,

Tomer Bitan: Um okay, maybe one last tiny question. Do you think it'll be horrible if we if we need to give up on the heckbench side of the data and only focus on the math one? Say the math one is like it's like a minute per per data point and the the other one is uh 10 minutes per data point. If I just have the math one and I do the entire process on math, is What?

Niranjan Hasabnis: Sorry, my son was shouting in the background, so I I apologize. Yeah.

Tomer Bitan: Yeah.

Niranjan Hasabnis: Was

Tomer Bitan: All good. So, so do you think it's acceptable or uh a a big

Niranjan Hasabnis: again

Tomer Bitan: like is it so bad if we have to like ditch the heckbench and focus on the math reasoning since it's so much faster to get results with that thing? Like do you think that's a no-go or so?

Niranjan Hasabnis: what? What is matter isn't it?

00:58:17

Tomer Bitan: So uh there's a a math shepardd it's called they made like a a GSM8K like

extension for this type of model that is like uh guiding a bigger model. Um and in order and like they have a huge data set and it's all like this

Niranjan Hasabnis: Okay.

Tomer Bitan: kind of like math like Tommy goes to the store and he has \$2 and he's it's like like this kind of math reasoning questions. Um, and so when I'm running on that, it takes me a minute to get like a data point so that I can train my tiny model on. But when I'm running on Hecbench,

Niranjan Hasabnis: Okay.

Tomer Bitan: it takes me 10 minutes to get a single data point.

Niranjan Hasabnis: I see.

Tomer Bitan: So I I was thinking

Niranjan Hasabnis: But this is this is for a different problem right this is a math not not

Tomer Bitan: like Yeah. Yeah. Yeah. Yeah.

Niranjan Hasabnis: uh parallel programming right yeah I mean you can actually evaluate whether the

Tomer Bitan: Yeah. Yeah.

Niranjan Hasabnis: idea works but I don't I don't know whether we can strongly say that

00:59:12

Tomer Bitan: write a full paper.

Niranjan Hasabnis: the model works on mass will work on parallel programming right or can

Tomer Bitan: Yeah.

Niranjan Hasabnis: we

Tomer Bitan: Yeah. Yeah. Yeah. Is it is it too hard of a pivot to say here is a reasoning technique that works

Niranjan Hasabnis: Um,

Tomer Bitan: on math. I want to like and then leave out the parallel program.

Niranjan Hasabnis: I mean we want to make

Tomer Bitan: It's a bit sad to leave it out the parallel programming though. So I don't know like

Niranjan Hasabnis: the paper specific to parallel programming right but yes if this is a more of a general technique that works across any domain then maybe we

Tomer Bitan: Mhm.

Niranjan Hasabnis: can write a paper that way and then think about parallel programming

Tomer Bitan: Okay.

Niranjan Hasabnis: right but yeah if you think that this evaluation is much faster to get

Tomer Bitan: Okay.

Niranjan Hasabnis: maybe within a day or so you can actually evaluate this then maybe we can try it So

Tomer Bitan: Okay. Yeah. Okay.

01:00:06

Tomer Bitan: So,

Niranjan Hasabnis: basically

Tomer Bitan: start start with don't no giving up on parallel codes yet. Start with trying as fast as I can to get some result on the math and see if I can get it to translate over to the co to the parallel codes. Right.

Niranjan Hasabnis: I I think overall I I think we first need to check that the idea itself works

Tomer Bitan: That's Yeah.

Niranjan Hasabnis: right.

Tomer Bitan: Yeah. Yeah.

Niranjan Hasabnis: So do you know that there is the that the idea work or is it just the intuition that you have and

Tomer Bitan: No.

Niranjan Hasabnis: then okay

Tomer Bitan: Yeah. It's mostly intuition and it's like extrapolating from other papers,

Niranjan Hasabnis: and

Tomer Bitan: but I haven't a full I don't have a full test showing its works

Niranjan Hasabnis: I say maybe it's worth checking that the idea itself works on a

Tomer Bitan: yet. Okay.

Niranjan Hasabnis: simpler benchmark maybe like a math reasoning if you want and then we can think about

Tomer Bitan: Mhm.

Niranjan Hasabnis: parallel programming reasoning as a next step right because if it it takes a

01:00:54

Tomer Bitan: Okay. Yeah.

Niranjan Hasabnis: long time and when the the idea does not work on math then I think there's a less chance it will work on

Tomer Bitan: Yeah. Yeah.

Niranjan Hasabnis: par programming okay yeah and we can actually validate

Tomer Bitan: Yeah, yeah, yeah. Okay. Yeah, I got you. Thank you. Yeah,

Niranjan Hasabnis: the idea itself on the math first and then go to par

Tomer Bitan: that's helpful.

Niranjan Hasabnis: programming

Tomer Bitan: Okay, awesome. Yeah, that actually helps clear some stuff in my head. So, thank

Niranjan Hasabnis: right no I think the idea is pretty interesting but I mean there I think we need to

Tomer Bitan: you.

Niranjan Hasabnis: try a few things to to verify that the idea works and then move to the more complex problem of parallel programming. Right.

Tomer Bitan: Yeah. Yeah. Yeah.

Niranjan Hasabnis: All right. I mean feel free to we can we set up a meeting again if you want with more details because I don't have the full like

Tomer Bitan: Okay.

Niranjan Hasabnis: context and understanding but if you want we can set up sometime next week and if you send me

Tomer Bitan: Okay. Okay.

Niranjan Hasabnis: your like I mean I think you don't have results so far but whatever you have done so far I think in terms of experiments maybe if you can share that we I will get more like understanding of the situation and then I can can talk

Tomer Bitan: Okay.

Niranjan Hasabnis: more about that.

Transcription ended after 01:02:18

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